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Original article

Potential genes associated with survival of *Acinetobacter baumannii* under ciprofloxacin stressShruti Kashyap^a, Prince Sharma^b, Neena Capalash^{a,*}^a Department of Biotechnology, Panjab University, BMS Block-I, Sector-25, Chandigarh, 160014, India^b Department of Microbiology, Panjab University, BMS Block-I, Sector-25, Chandigarh, 160014, India

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ABSTRACT

Acinetobacter baumannii is an opportunistic pathogen that has acquired resistance to all available drugs. The rise in multi-drug resistance in *A. baumannii* has been exacerbated by its ability to tolerate antibiotics due to the persister cells, which are phenotypic variants of normal cells that can survive various stress conditions, resulting in chronicity of infection. In the present study we observed that *A. baumannii* formed persister cells against lethal concentration of ciprofloxacin in exponential phase. The transcriptome of *A. baumannii* was analyzed after exposure to high concentration of ciprofloxacin (50X MIC) to determine the possible mechanisms of survival. Transcriptome analysis showed differential expression of 146 genes, of which 101 were up-regulated and 45 were down-regulated under ciprofloxacin stress. Differentially expressed genes that might be important for persistence against ciprofloxacin were involved in DNA repair, phenylacetic acid degradation, leucine catabolism, HicAB toxin-antitoxin system and ROS response (iron-sulfur clusters, hemerythrin-like metal binding and Kdp). *recA*, *umuD* and *ddrR* genes involved in SOS response were also up-regulated. Knockout of *umuD* showed significant decrease in persister cells formation while they were completely eradicated in *recA* mutant strain. The differentially expressed genes highlighted in the study merit further investigation as therapeutic targets for effective control of *A. baumannii* infections.

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Acinetobacter baumannii is a major nosocomial pathogen involved in various life-threatening diseases, including ventilator-associated pneumonia (VAP), meningitis, bacteraemia, and infections related to skin and soft tissue, central nervous system, urinary tract, and abdominal [1]. *A. baumannii* has developed resistance to a wide variety of antibiotics and recent studies have shown that the prevalence of multidrug-resistant *A. baumannii* is on the increase [2,3]. Carbapenem-resistant *A. baumannii* has been declared an urgent threat by the WHO (2017) and the CDC (2019), necessitating the development of new antibiotics. However, the resistant characteristic of *A. baumannii* is not the sole cause of therapeutic failure. The presence of persister cells could indeed be an important factor in the relapse of infections and chronic diseases [4]. Persisters are a subset of antibiotic-sensitive bacterial cells that can tolerate lethal concentrations of antibiotic. They are distinct

from resistant cells as they do not divide actively in the presence of antibiotics and do not show change in MIC after regrowth [5,6].

Ciprofloxacin is a fluoroquinolone antibiotic that acts by inhibiting the DNA gyrase and topoisomerase IV enzymes. Fluoroquinolone resistance involves chromosomal mutations in genes encoding the quinolone targets (DNA gyrase and topoisomerase IV) and plasmid-mediated quinolone resistance (PMQR) mechanism [7,8]. Quinolones have shown good activities against *Acinetobacter* spp. However, over the past three decades, resistance to these antibiotics in *A. baumannii* isolates has rapidly increased [9,10]. Besides this, persister formation in *A. baumannii* against ciprofloxacin has also been reported [11,12] but mechanism involved has not been elucidated.

Herein, to understand the response of *A. baumannii* to high concentration of ciprofloxacin, transcriptome analysis was performed. Our results underscore the importance of genes associated with SOS and ROS response, DNA repair, phenylacetic acid and leucine degradation, and HicAB toxin-antitoxin system for survival of *A. baumannii* under ciprofloxacin stress. These findings provide

* Corresponding author. Department of Biotechnology, South Campus, Basic Medical Science Block-I, Panjab University, Sector-25, Chandigarh, 160014, India.

E-mail address: caplash@pu.ac.in (N. Capalash).

new target genes that can be studied further for their role in *A. baumannii* persistence and as potential therapeutic targets.

1. Materials and methods

1.1. Bacterial strains, media and growth conditions

A. baumannii ATCC 17978 strain was procured from the American Type Culture Collection (ATCC, Manassas, VA, USA). *A. baumannii* ATCC 17978 *recA* and *umuD* deletion mutants were gift from Dr. German Bou Arevalo from the Department of Microbiology and Parasitology, Complejo Hospitalario Universitario La Coruña, Spain. *A. baumannii* strains used in the study were routinely grown and maintained in Lysogeny Broth (LB).

1.2. Minimum inhibitory concentration (MIC)

The MIC was determined using the broth microdilution method as per the CLSI guidelines [13]. Ciprofloxacin was serially diluted from 128 to 0.25 µg/ml. The MIC was described as the concentration at which no visible growth occurred.

1.3. Persister assays

The overnight grown culture was inoculated (1:100) in fresh LB broth and grown to exponential phase ($OD_{600} \sim 0.5$) at 37 °C, 180 rpm. The concentration- and time-dependent persister assays were performed to determine colony-forming units per ml (CFU/ml) and percent survival as described previously [12].

1.4. RNA isolation, cDNA library construction, and transcriptome analysis

1.0 ml of exponential phase cells ($OD_{600} \sim 0.5$) were treated with 50X MIC of ciprofloxacin (12.5 µg/ml) at 37 °C, 180 rpm for 60 min. Total RNA was isolated by combining TRI Reagent (Sigma) with RNeasy Mini Kit (Qiagen, USA) according to the manufacturer's instructions. The RNA samples were treated with Turbo DNase I (Invitrogen) and the purity was checked on 1% agarose gels for evaluating the 23S and 16S ribosomal RNA bands (23S/16S ratio). RNA was quantified using the NanoDrop (Eppendorf) and the samples with $OD_{260/280}$ ratio greater than 1.9 were sequenced by outsourcing to SciGenom Labs Pvt. Ltd. (Kochi), India. RNA integrity number (RIN) of samples was determined on Bioanalyzer (Agilent 2100, Agilent Technologies, USA) and samples with a RIN > 7 were used for cDNA library construction. All cDNA libraries were sequenced using Illumina HiSeq 2500 platform and paired-end strategy (read length 100 bp). The obtained RNA-Seq reads were aligned with *A. baumannii* ATCC 17978 genome using Hisat2 [14]. Differential gene expression analysis was performed using Cuffdiff [15]. The genes with an adjusted *p* value ≤ 0.05 and $|\log_2 FC| \geq 2$ were designated as significant differentially expressed genes (DEGs).

1.5. Functional annotation of differentially expressed genes

Functional analysis of the identified genes was performed through Gene Ontology (GO) annotation using the OmicsBox version 3.2. Similarity search of predicted genes was performed by NCBI blast + via CloudBlast on BLASTX program with non-redundant protein sequences (Nr_v5) and UniProtKB/Swiss-Prot (swissprot_v5) databases. The genes were further classified using the Kyoto Encyclopedia of Genes and Genomes (KEGG) metabolic pathways database and Cluster of Orthologous Groups (COGs) protein database.

1.6. Validation of data by qRT-PCR

To validate the transcriptome data, qRT-PCR was performed using the RNA extracted under similar conditions as that for RNA-Seq analysis. cDNA was synthesized using the verso cDNA synthesis kit (Thermo Fisher, Co., USA) according to the manufacturer's instructions. Primers were designed using Primer3 software and 16S rRNA gene was used as the housekeeping control. All qRT-PCR reactions were run on CFX96 Touch Real-Time PCR detection system (BioRad, USA) using the SYBR Green Master Mix according to the manufacturer's protocol. qRT-PCR was performed with three biological replicates for each sample. The relative expression of each gene was determined by the $2^{-\Delta\Delta Ct}$ method [16].

1.7. Protein–protein interaction (PPI) network analysis

STRING (version 11.0) was used to construct the PPI network of DEGs, with a combined score > 0.7 (high confidence) as the cut-off criterion. An open platform, Cytoscape (version 3.8.2), was used to visualize molecular interaction networks based on the results from STRING. The CytoHubba plug-in of Cytoscape [17] identified the top 10 hub genes in the network with the highest degree of connectivity.

2. Results

2.1. Persister cells formation by ciprofloxacin treatment

To determine the persister subpopulation, antibiotic-killing assay was performed by exposing exponential-phase *A. baumannii* cells to increasing concentrations of ciprofloxacin. The biphasic death curve showed an initial rapid killing of the majority of cells levelling off at 50X MIC (MIC = 0.25 µg/ml) followed by a plateau of surviving persister cells (Fig. 1A). The time-

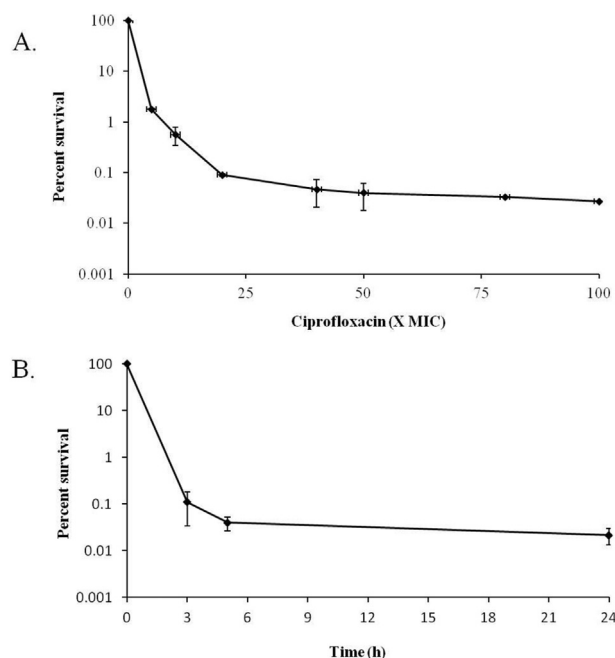


Fig. 1. Exponential phase cells of *A. baumannii* ATCC 17978 on treatment with (A) different concentrations of ciprofloxacin for 3 h and (B) 50X ciprofloxacin for different time intervals. Untreated cells at 0 h ($3.2 \pm 0.36 \times 10^8$ CFU/ml) were taken as 100% and percent survival was calculated accordingly at different concentrations of ciprofloxacin (A) and different treatment duration (B). Data are represented as mean log percent survival $\pm$ SD of three independent experiments.

dependent assay with 50X MIC of ciprofloxacin revealed the presence of 0.02% persister cells at 24 h (Fig. 1B).

2.2. Global analysis of differentially expressed genes (DEGs) in response to ciprofloxacin treatment

A total of 146 genes were differentially expressed, including 101 up-regulated and 45 down-regulated ones, in response to high concentration of ciprofloxacin (50X MIC). The number of up-regulated genes was found to be larger than that of the down-regulated genes. GO analysis showed a set of 95 DEGs (65%) with known annotations. The largest group of 50 DEGs (37 up and 13 down), was involved in metabolic processes. Majority of the proteins encoded by these DEGs were intrinsic component of membrane (14 up and 8 down) and their molecular functions were primarily related to “catalytic activity” (41 up and 15 down) and “binding” (28 up and 14 down) (Fig. 2A). COG analysis revealed that a large number of up-regulated genes were associated with lipid transport and metabolism (COG I) and replication, recombination and repair (COG L) while the majority of down-regulated genes were linked to transcription (COG K) and inorganic ion transport and metabolism (COG P) (Fig. 2B). KEGG analysis classified 20 DEGs to 49 different pathways and most were associated with amino acid and carbohydrate metabolism (Fig. 2C).

It was observed that there were 35 DEGs (including 26 up-regulated and 9 down-regulated) that were highly up- and down-regulated ($\log_2FC < -4$ or > 4) (Table S1). A few prominent up-regulated ones were those involved in DNA repair (*ddrR*, A1S_1388; *umuD*, A1S_1389), methylation (*dcm*, A1S_2036; *rlmD*, A1S_1581), leucine degradation pathway (*liuC*, A1S_1374; *mccB*, A1S_1375; *ivd*, A1S_1376) and HicAB toxin-antitoxin system (*hicB*, A1S_2019). Similarly, hemerythrin-like metal-binding protein (A1S_0891), cold shock protein (*cspV*, A1S_1228), rare lipA family protein (*rlpA*, A1S_2317) and D-amino acid dehydrogenase (*dadA*, A1S_1075) were highly down-regulated in response to ciprofloxacin treatment.

The rest of the 111 DEGs (including 75 up-regulated and 36 down-regulated) showed change in the expression by 4-fold ($\log_2FC < -2$ or > 2) (Table S2). Genes involved in the phenylacetic acid catabolism (A1S_1335, *paaN*; A1S_1337, *paaB*; A1S_1338, *paaC*; A1S_1339, *paaD*; A1S_1340, *paaE*; A1S_1343, *paaH*), production of acetyl-CoA (*ackA*, A1S_0482; *pta*, A1S_0481) and DNA replication, recombination and repair (*parE*, A1S_3359; *recA*, A1S_1962; *umuC*, A1S_2008; *umuD*, A1S_1174; *ddrR*, A1S_1388; *umuD*, A1S_1389; *ruvB*, A1S_2588; *recB*, A1S_0356; *repA*, A1S_0663) were up-regulated. The virulence-associated toxin (*hicA*, A1S_2020), outer membrane lipoprotein carrier protein (*lolA*, A1S_2729), potassium-transporting ATPase (*kdpC*, A1S_2139) and iron-sulfur cluster binding protein (*iscX*, A1S_0497; *iscA*, A1S_1630) were also among the overexpressed genes. Ciprofloxacin, on the other hand, reduced the expression of genes involved in sulfur metabolism (*cysN*, A1S_1001; *cysD*, A1S_1000) and copper resistance protein (*copD*, A1S_2941).

2.3. Validation of transcriptome data by qRT-PCR

RNA-Seq data was confirmed by qRT-PCR analysis of six DEGs involved in phenylacetic acid metabolism (*paaE*), methylation (*dcm*), SOS response (*recA*), toxin-antitoxin system (*hicA* and *hicB*) and outer membrane protein (A1S_3297), using gene-specific primers (Table S3). Treatment with 50X ciprofloxacin significantly increased the expression of *paaE*, *dcm*, *recA*, *hicA* and *hicB* by 5.34, 102.65, 3.14, 2.34 and 1.53-fold respectively and down-regulated the expression of gene A1S_3297 (66.6-fold) encoding for outer membrane beta-barrel domain protein (Fig. 3). The expression

patterns of six genes determined by qRT-PCR were found to be consistent with the transcriptome data.

2.4. PPI network analysis

The PPI network analysis to obtain insights into the functional interaction of the DEGs revealed 39 nodes (including 32 up-regulated and 7 down-regulated genes) and 37 interactions (Fig. S1). The top 10 genes with the highest degree of connections were identified as *paaE*, *paaD*, *paaC*, *paaB*, *paaH*, *paaN*, *mccB*, *umuD*, *recA* and *pta* (Table S4), which could be considered as the hub genes.

2.5. Role of SOS response in *A. baumannii* survival against ciprofloxacin

Involvement of SOS response genes, *recA* (A1S_1962) and *umuD* homolog (A1S_1389) (also identified as hub genes in PPI network analysis) was evaluated for survival against ciprofloxacin (50X MIC) using *recA* and *umuD* mutant strains. The growth rate of *A. baumannii* *recA* and *umuD* deletion mutants was similar to that of the wild-type (Fig. 4A). There was no change in the MIC of ciprofloxacin for *umuD* mutant (MIC, 0.25 µg/ml) but *recA* mutant showed a two-fold decrease (MIC, 0.125 µg/ml) as compared to the wild-type. Treatment of wild-type with 50X MIC of ciprofloxacin for 1 h showed 3.3% cell survival which was reduced by 550- and 3.5-fold in *recA* and *umuD* deletion mutants respectively. After 3 h, 0.17% cells survived in wild-type while there was 4-fold reduction in *umuD* mutant and eradication of persister cells in *recA* mutant (Fig. 4B).

3. Discussion

Besides the global problem of antibiotic resistance, antibiotic failure due to persister cells further complicates the treatment of *A. baumannii* infections [18,19]. In this study, *A. baumannii* cells in exponential phase were found to form persister cells against ciprofloxacin (a gyrase inhibitor). Previous studies have shown persister cells formation in different growth phases of *A. baumannii* against ciprofloxacin [11,12] but the mechanism(s) involved has not been explained. Thus to gain insight into the genes that may be critical for survival and persistence against ciprofloxacin, we analyzed the transcriptome of *A. baumannii* cells after exposure to high concentration (50X MIC) of ciprofloxacin which killed the normal population as per the biphasic killing curve.

In the present study, genes encoding DNA cytosine methyltransferase and putative methyltransferase were highly expressed in response to ciprofloxacin. A previous study has also shown induction of *dcm* on treatment with DNA damaging agent mitomycin C in *A. baumannii* [20]. *Dcm* has been shown to repress the expression of stationary phase sigma factor *rpoS* in *E. coli* [21] and also influenced drug resistance through increased expression of drug transporter *SugE* [22]. In *E. coli*, deletion of *dam* caused significant decrease in persister formation during exposure to different classes of antibiotics (gentamicin, fluoroquinolones and cephalosporins) [23]. DNA methylation is the primary means of epigenetic gene regulation in bacteria [24]. Hence, role of DNA methylation in persistence of *A. baumannii* against stress conditions needs to be investigated further.

Ciprofloxacin is known to cause DNA damage which triggers SOS response [20], with *RecA* and *LexA* as the key regulators [25]. Previous studies have also shown that mutations in *recA* and *recB* caused decrease in persister survival when challenged with ciprofloxacin [26] and were necessary for persister survival against ofloxacin treatment in *Escherichia coli* [27]. Homolog of *LexA*

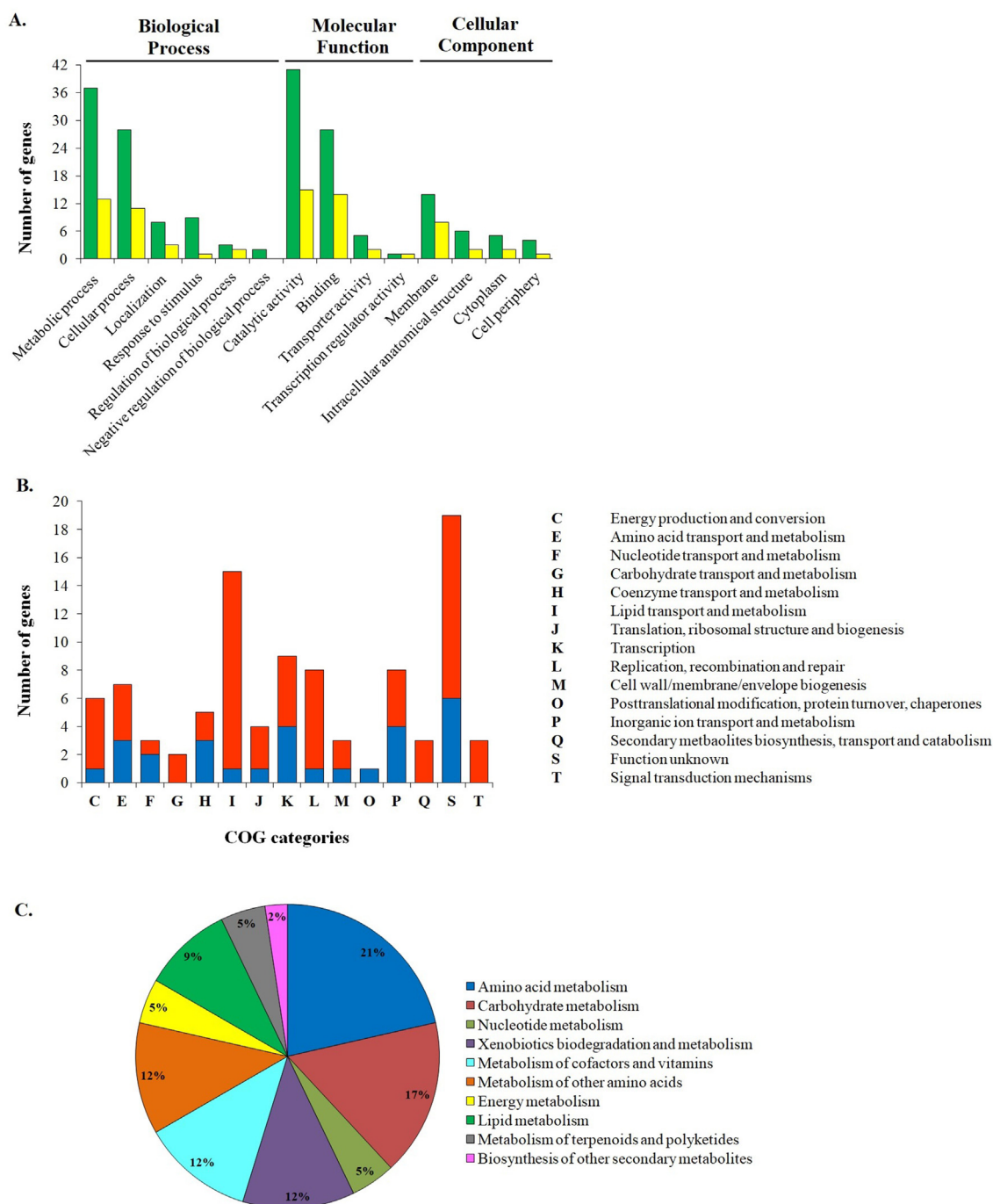


Fig. 2. (A) Functional categories of differentially expressed genes (DEGs) based on Gene ontology classification. The green and yellow boxes indicate up-regulated and down-regulated genes, respectively. (B) Cluster of Orthologous Groups of proteins (COG) analysis of DEGs. The orange and blue boxes indicate up-regulated and down-regulated genes, respectively. (C) Pie chart showing KEGG pathways analysis of DEGs.

repressor has not been identified in *Acinetobacter* species while UmuD homologue acts as an indirect regulator of DNA damage response in *A. baumannii* [28]. DdrR is UmuD-regulated DNA damage inducible protein that is exclusive to *Acinetobacter* genus, however its function is unknown [29]. In the present study, there was significant up-regulation of genes involved in SOS response, including *recA*, *umuC*, *umuD* and *ddrR*, and other genes involved in DNA repair (*ruvB*, *recB* and *repA*). Significant reduction in the survival of *umuD* mutant and complete eradication of persister cells in

recA mutant proved their importance in survival of *A. baumannii* against ciprofloxacin.

The phenylacetic acid (PAA) catabolic pathway, encoded by the *paa* operon is the central route for the metabolism of many aromatic compounds that finally converges in tricarboxylic acid cycle [30]. PAA catabolic pathway has been shown to be associated with virulence in *A. baumannii* [31]. Deletion of *paaA* resulted in the accumulation of phenylacetate, which served as a neutrophil chemo-attractant, resulting in a decrease in the severity of

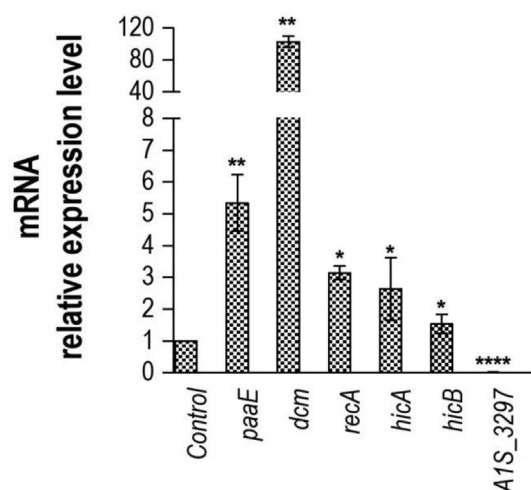


Fig. 3. Relative expression of *paaE*, *dcm*, *recA*, *hicA*, *hicB* and A1S_3297 genes in exponential phase cells of *A. baumannii* ATCC 17978 upon treatment with ciprofloxacin (50X MIC). Untreated cells were taken as the control with basal level expression indicated as 1. Data are representative of three independent experiments. Bars represent the mean $\pm$ SD. * $P \leq 0.05$; ** $P \leq 0.01$; **** $P \leq 0.0001$.

A. baumannii infection in a zebrafish model [32]. Increased expression of *paa* operon genes has also been observed in *A. baumannii* persister cells formed in response to ceftazidime, a third generation cephalosporin [33]. The present study also supports the involvement of PAA catabolism in persistence against ciprofloxacin, another class of antibiotic. Further studies are needed to establish the role of *paa* operon in *A. baumannii* against different stress conditions.

Amino acids have been shown to play significant role in persister tolerance [34,35]. Activation of genes for the synthesis of serine, aspartate, phenylalanine and tyrosine has been shown to contribute to persister formation in *E. coli* against gentamicin [36]. Leucine acts an alternative carbon source required for metabolism during starvation and has been shown to increase the chances of survival of *Staphylococcus xylosus* [37]. Leucine is catabolised to acetyl-CoA which fuels TCA cycle [38]. Three genes encoding key enzymes viz. methylglutaconyl-CoA hydratase, 3-methylcrotonyl-CoA carboxylase beta subunit and isovaleryl-CoA dehydrogenase, involved in leucine degradation pathway were up-regulated in response to ciprofloxacin. In addition, ciprofloxacin up-regulated the expression of acetate kinase (*ackA*) and phosphate

acetyltransferase (*pta*), enzymes involved in the AckA-Pta pathway resulting in synthesis of acetyl-CoA [39]. Deletion of either *pta* or *ackA* caused significant reduction in the *Staphylococcus aureus* growth [40]. These pathways forming acetyl-CoA may be important to meet the energy requirements for survival under ciprofloxacin stress.

Toxin-antitoxin modules play an important role in bacterial physiology [41]. The HicAB system, which belongs to the type II TA systems, has been implicated in stress response, virulence and persistence [42,43]. Deletion of *hicAB* was found to reduce persister formation in *Burkholderia pseudomallaei* against ciprofloxacin [44] and *Acetobacter pasteurianus* against kanamycin [45]. Various type II TA systems has been identified in *A. baumannii* [46] but their role in persistence has not been elucidated. Increased expression of *hicA* and *hicB* genes in response to high concentration of ciprofloxacin suggested their role in survival which requires further investigation.

The lipoprotein outer membrane localization (Lol) system plays a central role in the sorting of proteins to the outer membrane [47]. In *E. coli*, Lol system is considered essential for cell growth [48]. Increased expression of *lolA* may be required for stabilization of damaged outer membrane in colistin treated *A. baumannii* [49], however its role in ciprofloxacin stress needs elucidation.

Ciprofloxacin also increased the expression of the *kdpC* gene, which is a part of the inner membrane potassium ion transporting ATPase complex regulated by KdpD/KdpE two-component system (TCS) [50]. Potassium transport (Kdp) system is well conserved in *A. baumannii* strains and has been reported to have role in pathogenesis [51]. Involvement of Kdp operon in resistance to antimicrobial and oxidative stress in intracellular pathogens has also been reported [52]. Since ciprofloxacin also generates reactive oxygen species (Fig. S2) in *A. baumannii*, *kdpC* could have a protective role.

Iron-sulfur (Fe-S) clusters are important source of iron for Fenton reaction-mediated hydroxyl radical formation and serve as cellular redox switches that help cells recover from stressful conditions [53]. Deletion of *iscR* decreased virulence of *Pseudomonas aeruginosa* in *Drosophila melanogaster* and mouse peritonitis models [54]. Up-regulation of *iscA* and *iscX* genes that are part of iron-sulfur cluster (ISC) assembly system might be a response to oxidative stress of ciprofloxacin.

Few genes were highly down-regulated in response to ciprofloxacin of which hemerythrin-like iron and oxygen binding protein showed maximum down-regulation. In addition to acting as an oxygen sensor in bacteria, they can store and transport oxygen [55]. Knockout of hemerythrin-like protein in *Mycobacterium smegmatis*

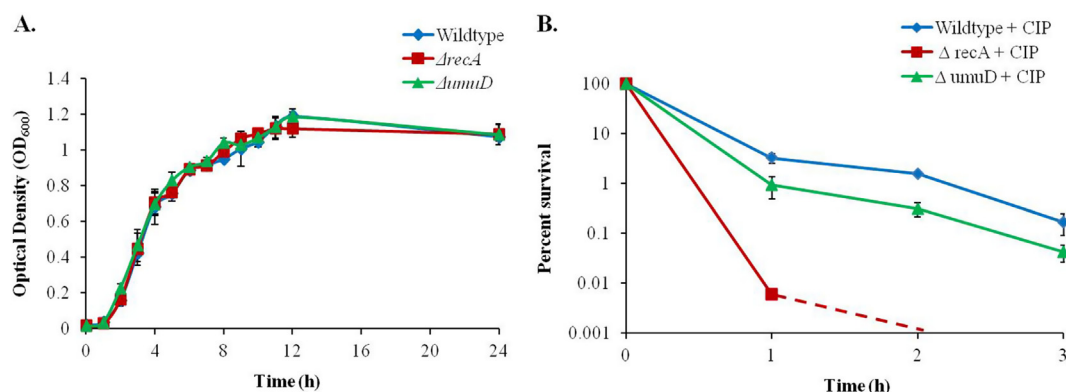


Fig. 4. (A) Growth of *A. baumannii* ATCC 17978, and $\Delta recA$ and $\Delta umuD$ mutants in lysogeny broth. (B) Exponential phase cells were treated with 50X ciprofloxacin for different time intervals. 100% CFU/ml at 0 h were $8.3 \pm 0.26 \times 10^8$ for ATCC 17978, $2.82 \pm 0.15 \times 10^8$ for $\Delta recA$ and $2.22 \pm 0.4 \times 10^8$ for $\Delta umuD$. Data are represented as mean log percent survival $\pm$ SD of three independent experiments.

was resistant to hydrogen peroxide stress [56] which underscored its role in response to oxidative stress. Similar function of hemerythrin-like protein is expected in *A. baumannii* and its down-regulation might have a protective role. Cold shock proteins (CSPs) have been shown to be important for bacterial adaptation and intracellular survival in response to different stress conditions [57–60]. Deletion of CSP family (CspA, CspB and CspD) proteins in *Listeria monocytogenes* led to increased susceptibility to oxidative stress and impaired invasion of the host cells and intracellular growth [61]. Since *cspV* was significantly down-regulated in *A. baumannii* under ciprofloxacin stress, it is important to study its targets to understand its role in persistence. D-amino acid dehydrogenase (DadA) is an iron-sulfur flavoprotein that mediates the oxidation of D-amino acids into their corresponding keto-acids [62] which undergo decarboxylation in response to oxidative stress and help in ROS detoxification [63]. Since *dadA* was significantly down-regulated in *A. baumannii* its role in survival against ciprofloxacin could not be defined.

Taken together, the present study shows the predominance of up-regulated genes than down-regulated ones in response to ciprofloxacin stress. Since ciprofloxacin causes DNA damage, hence genes associated with SOS response and DNA repair were up-regulated for survival in *A. baumannii*. Also, phenylacetic acid and leucine degradation provided the energy requirements for survival under ciprofloxacin stress. In addition, Kdp system, iron-sulfur cluster and hemerythrin-like protein might be important in attenuating oxidative stress generated by ciprofloxacin. This study also shows that many genes previously linked to virulence were associated with survival under ciprofloxacin stress which merit further investigation for effective control of *A. baumannii* infections.

Declaration of competing interest

None of the authors have any conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micinf.2021.104844>.

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